Amazon Product Data Analysis using SQL

Extracting actionable insights from pricing, discounts, ratings, and customer reviews to drive e-commerce decisions

Made with GAMMA

Dataset Overview

Product-Level Data

Product names, categories, pricing details, and discount information

Review-Level Data

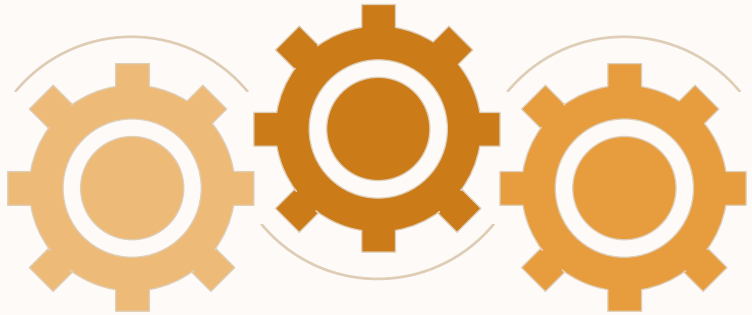
Customer ratings, review counts, and detailed feedback content

Key Metrics

Actual price, discounted price, discount percentage, and rating scores

Result Set

Filtered records derived from the original source table.



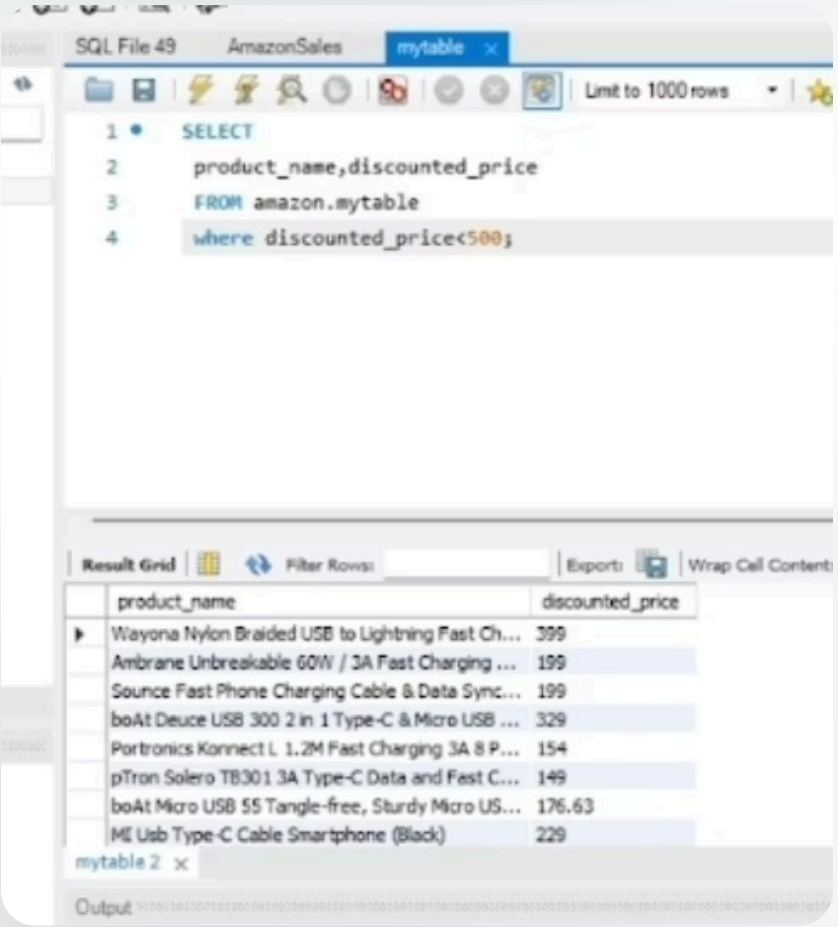
Source Table

The initial collection of data before any filtering is applied.

Final Data Point

The core outcome or single record from the process.

Made with GAMMA



Key SQL Concepts Applied

1

Filtering Operations

WHERE, BETWEEN, and LIKE clauses for precise data selection

2

Aggregation Functions

AVG() for calculating average ratings and pricing metrics

3

String Matching

LIKE operator with wildcards (%) for pattern-based searches

4

Logical Operators

AND, OR for combining multiple conditions in queries

Made with GAMMA

Business Problems Solved

01

Identify Budget Products

Find products priced below ₹500 for value-conscious customers

02

Detect High-Discount Items

Flag products with 50% or more discount for promotional analysis

03

Analyze Mid-Range Offers

Study products with 20-40% discounts for pricing strategy insights

04

Spot Premium Products

Locate high-rated products for quality benchmarking

05

Extract Customer Sentiment

Analyze review content for positive and negative feedback patterns

06

Category Analysis

Identify product categories, particularly Accessories segment using query

Query Intent

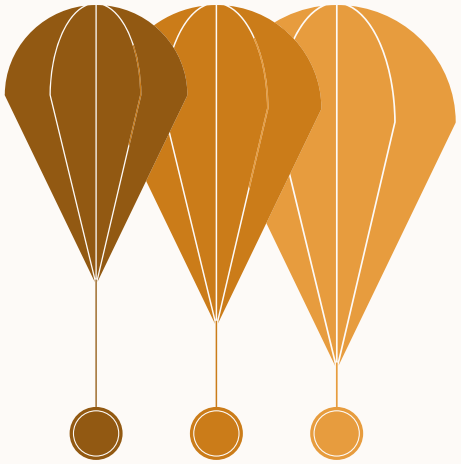
Retrieve product names and categories

Filter Criteria

Category contains 'Accessories'

Result Set

List of matching products



Made with GAMMA

Advanced Query Techniques

Price Comparison

Calculate actual versus discounted price differences to measure discount effectiveness and identify pricing anomalies in the dataset

example

```
SELECT product_name,discount_percentage FROM amazon.mytable where discounted_price like ("%9");
```

Price comparison

Discount	Pricee Price	Priced Discount	Wäten $1.00	Caoral $24.0um
Unals	✓	✓	✓	✓
Descuin	✓	✓	✓	✓
Discound	✓	✓	✓	✓
Eola	✓	✓	✓	✓
Ro%	✓	✓	✓	✓
Say%	10%	50%	1.0%	5.0%

Pattern-Based Filtering

Product names containing "Cable"

Price Pattern Matching

Prices ending with digit 9

Text Analytics

Keyword extraction from reviews

Made with GAMMA

Key Findings

Category Dominance

Accessories category represents the largest segment of products in the dataset

High-Discount Products

Cable products frequently feature discounts of 50% or higher

High-Discount Products

Cable products frequently feature discounts of 50% or higher

High-Discount Products

Cable products frequently feature discounts of 50% or higher

Positive Feedback

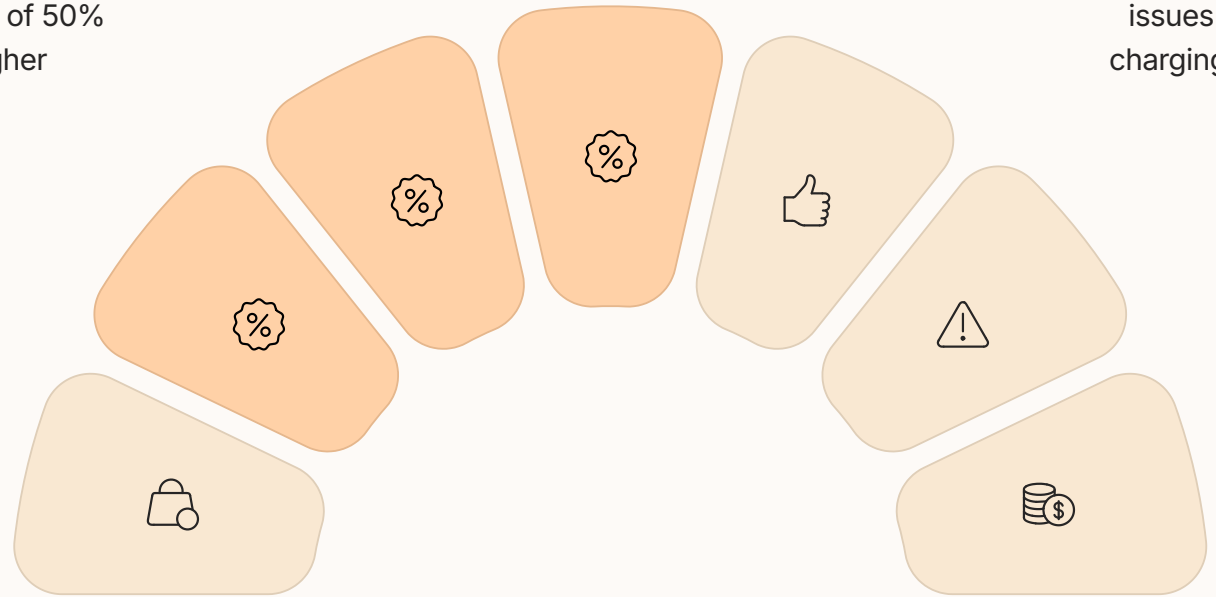
Customers praise fast charging capabilities and product durability

Quality Concerns

Negative reviews mention quality issues and slow charging problems

Budget Market

Affordable products dominate listings with strong customer interest



Made with GAMMA

Use Cases

E-commerce Product Analysis

Comprehensive evaluation of product portfolios across categories and price points

Pricing Strategy Optimization

Data-driven insights for competitive pricing and promotional campaign planning

Customer Sentiment Analysis

Understanding customer preferences and pain points through review mining

Product Recommendation Systems

Building intelligent systems based on pricing, ratings, and customer feedback

Made with GAMMA

Tools & Technologies

SQL Engine

MySQL for executing analytical queries on product data

Amazon Product Dataset

e-commerce data including pricing, reviews, and ratings

Query-Based Analysis

Transforming raw data into business insights through structured queries

Made with GAMMA

Learning Outcomes

1 Enhanced SQL Skills

Master filtering, aggregation, string matching, and logical operations for complex queries

2 Real-World Dataset Experience

Work with authentic e-commerce data representing actual business scenarios

3 Practical Applications

Hands-on practice in data filtering, text analysis, and business insight generation

Made with GAMMA

Conclusion

SQL serves as a powerful tool for extracting meaningful insights from raw e-commerce data, enabling businesses to make data-driven decisions. This analysis demonstrates how structured queries can reveal pricing patterns, customer sentiment, and market trends.

By combining filtering techniques, aggregation functions, and text analysis, analysts can transform product data into actionable business intelligence. The ability to understand pricing strategies and customer behavior through reviews provides a competitive advantage in e-commerce.

Made with GAMMA

Created by -SHREYA KALLA

link -https://github.com/shreyakalla17-cell/amazon-sales.git

Made with GAMMA